

$$= Z[A] \exp(2\pi i \int_Y K F)$$

$$\Rightarrow 2k+1 + K = -K$$

$$K = -\frac{1}{2}(2k+1)$$

$$\Rightarrow \tilde{Z}[\tilde{A}] = Z[A] \exp\left(-\frac{i}{2}(2k+1) \int_Y F\right)$$

Thus

$$\tilde{Z}[\tilde{A}] \text{ is } C\text{-invariant!} \quad \tilde{Z}[\tilde{A}^c] = \tilde{Z}[\tilde{A}]$$

Note: $Z[A] \rightarrow 1\text{dim anomalous theory.}$

$\tilde{Z}[\tilde{A}] \rightarrow 2\text{dim Anomaly free theory.}$

TODOs

- Compute deformation class $[u_\mu]$ of "particle on a circle."
- understand anomaly inflow picture for discrete symmetry $\mathbb{Z}_2 \times \mathbb{Z}_2$
 e.g. $\hookrightarrow$ turned on by adding $\cos(2x)$ to Lagrangian $\Rightarrow$ expect to be in the same deformation class as above.
 shift symmetry broken to discrete gauge group $\downarrow$
- RG flow of "particle on a circle."
- What if? Gauge C & consider shift symmetries? [repeat above analysis]